Fitting Person Detection on a Plain Microcontroller

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Code, data, full paper: `https://github.com/akash-reddy-k/edgeAi`

Commercial CCTV analytics (Hikvision AcuSense, Ambarella CV7) run person detection on purpose-built AI silicon with dedicated neural accelerators. This project asks how far equivalent functionality—loitering detection on a fixed-mount camera—can be pushed onto a **plain microcontroller with no such accelerator**, and measures the answer rather than assuming it.

Method. Fine-tune YOLOv8n for single-class person detection (ShanghaiTech Campus dataset), track with ByteTrack, export to ONNX, apply INT8 post-training quantization. An ONNX-graph profiler then computes flash and activation-RAM footprint directly—no vendor toolchain or account required, fully reproducible from committed code.

Finding 1 — quantization alone is not enough. The INT8 model is 72% smaller than its FP32 export for a 0.2% accuracy cost, but at the stock YOLOv8n architecture, **nothing fits any plain STM32, at any input resolution**—flash alone overshoots the largest plain part (STM32H743, 2 MB) by 50%.

Finding 2 — that is an architecture fact, not a hardware fact. Profiling shows YOLOv8n's large-object detection pathway (P5/32) accounts for 51.8% of parameters and is functionally unused in fixed-mount overhead CCTV, where a person never fills half the frame. Removing it cuts flash 2.8×—more than quantization achieved on its own.

Finding 3 — the fix does not cost accuracy, it improves it. Reinvesting that freed budget as extra width (rather than shrinking every scale uniformly) wins on every metric in a controlled, random-init-matched comparison, including +3.3 points of mAP50-95. The resulting architecture fits an STM32H743 at 320×320 with roughly 50% headroom on both flash and RAM—no NPU, no external memory.

Honest caveat. The accuracy numbers above are a relative comparison on a held-constant, self-referential label set (no ground-truth bounding-box annotations exist for this dataset); the MCU fit numbers are an analytical lower bound, not yet verified on physical hardware. Both limitations are stated explicitly in the full paper, and both are active next steps.

Why this, not a bigger model. The research contribution is the compression-and-deployment pipeline and its diagnostic method—distinguish what a flash budget is actually spent on before assuming quantization is the only lever—not a new detector.

Full paper (12 pages, arXiv-style): `writeup/paper.pdf` in the repo above.